

TED UNIVERSITY

CMPE491 Senior Project

BioAuth Project Specifications Report

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1. Introduction

Modern digital onboarding and authentication systems depend a lot on liveness detection methods that are based on visual behavior, like blinking or moving your head. But because generative AI and deepfake technologies are growing so quickly, attackers can make very realistic fake facial animations that can easily get around these standard verification systems. This vulnerability reveals a significant constraint: visual behavior alone is no longer an adequate indicator of human presence, necessitating authentication mechanisms that utilize inherent physiological signals.

The BioAuth project wants to close this security hole by making a biometric authentication system that can tell if someone is alive. BioAuth adds remote photoplethysmography (rPPG) to traditional ways of checking faces to make them better. Instead of relying on visible actions that can be easily faked, the system gets and looks at small changes in skin color that happen because blood flows through the face.

The BioAuth project suggests a biometric authentication system that is aware of physiological liveness to fix this security hole. BioAuth improves traditional facial verification methods by adding remote photoplethysmography (rPPG). The system doesn't use visible actions that can be easily faked. Instead, it extracts and analyzes small changes in skin color caused by blood flow directly from facial areas of interest in real time. The system uses advanced extraction algorithms like CHROM and POS to figure out if the identity shown belongs to a real person.

The proposed hybrid pipeline combines facial recognition, motion-based validation, and the study of physiological signals. BioAuth wants to make its system much more resistant to modern spoofing methods, such as print, replay, and AI-generated portrait animation attacks, by using a fusion-based decision-making system.

This document outlines the project specifications of the BioAuth system, detailing the project description, system constraints, ethical and professional considerations, and the formal functional and non-functional requirements necessary for implementation.

1.1 Description

The goal of BioAuth, a physiological liveness-aware biometric authentication system, is to increase the accuracy of remote identity verification. Incorporating remote photoplethysmography (rPPG) into the verification process, it mitigates the shortcomings of traditional face-based authentication.

Conventional authentication processes usually rely on visual behavior checks (such as blinking or head movement) to support facial similarity. These techniques prevent simple spoofing attempts, but they are very vulnerable to more complex attacks like deepfake injections and AI-generated portrait animations. By assessing intrinsic biological activity—more especially, the minute variations in facial skin color brought on by blood circulation—which are difficult for conventional displays to replicate or synthesize, BioAuth mitigates these risks.

The system uses a multi-stage pipeline to process camera input in real-time. In order to isolate particular Regions of Interest (ROI) appropriate for pulse extraction, it first detects faces and extracts facial landmarks. The system suppresses motion and illumination artifacts while extracting physiological signals from these isolated regions using sophisticated rPPG algorithms like CHROM and POS.

BioAuth employs a fusion-based decision mechanism instead of depending on a single metric. It assesses the quality of the extracted pulse signal, structural motion consistency, and facial similarity. The system divides the authentication attempt into three distinct states based on this multi-factor evaluation: Accept (real and active user), Reject (spoof attempt or mismatched identity), or Retry (insufficient signal quality due to environmental factors). Resistance to print, replay, and AI-driven manipulation attacks is greatly increased by this hybrid strategy.

1.2 Constraints

Numerous limitations arising from technical, environmental, economic, and ethical considerations affect the development and implementation of the BioAuth system. To make sure the system is practical, dependable, and accountable, these limitations must be considered.

1.2.1 Technical Constraints

The quality of the input data has a significant impact on the system's performance. Lighting, camera resolution, frame rate, and user movement are some of the variables that affect physiological signal extraction using rPPG. Excessive motion or poor video input can seriously impair signal reliability and impact authentication accuracy. Furthermore, computational complexity and system latency are constrained by real-time processing requirements.

1.2.2 Economic Constraints

The system should function with readily accessible hardware like standard webcams or mobile device cameras. This condition restricts the utilization of specialized sensors or expensive equipment. Furthermore, development needs to be executed within the limitations of the available computational resources and time since the project is confined to an academic semester.

1.2.3 Environmental Constraints

Environmental factors, especially changes in lighting, influence the success of extracting physiological signals. Inadequate lighting or fluctuating illumination can compromise the precision of analysis based on rPPG. Consequently, the system must function effectively in different real-world scenarios while ensuring satisfactory performance.

1.2.4 Social Constraints

Different user groups with different physical qualities, such as variations in skin tone, facial features, and environmental histories, must be able to operate the system. To avoid biased results or uneven performance among various groups, it is crucial to ensure fairness and inclusivity.

1.2.5 Political and Regulatory Constraints

BioAuth naturally manages sensitive biometric data since it processes physiological signals and face imaging. As a result, the system must adhere to all applicable data protection rules, particularly the General Data Protection Regulation (GDPR) globally and the Personal Data Protection Law (KVKK) in Turkey. The architecture must guarantee that biometric information is processed momentarily in volatile memory and is never transferred or retained permanently without the express, informed consent of the user.

1.2.6 Ethical Constraints

Sensitive personal information is involved in biometric authentication systems, which raises ethical issues. The system must guarantee that user data is not misused, kept, or processed without authorization. Furthermore, testing needs to be done in controlled settings without focusing on or trying to take advantage of real-world systems, including financial platforms.

1.2.7 Health and Safety Constraints

Because BioAuth is a purely software-based biometric authentication solution that uses common optical sensors (such as commercial webcams and smartphone cameras), it poses no direct risks to the user's physical health or safety. The device doesn't require physical touch or release any dangerous radiation while operating in a non-intrusive manner.

1.2.8 Manufacturability Constraints

Conventional physical manufacturing and assembly limitations do not apply to BioAuth because it is a software product. Rather, this project's "manufacturability" refers to cross-platform interoperability, deployment viability, and software development procedures. The system must be easily packageable for deployment on common commercial hardware without the need for specialist biometric sensors, and it must be constructed using accessible libraries (e.g., OpenCV, MediaPipe).

1.2.9 Sustainability Constraints

Both rPPG signal extraction and continuous real-time video processing are computationally demanding processes. In order to reduce CPU/GPU overhead, the system architecture must be optimized. Since high computational demand is strongly correlated with excessive power consumption and quick battery drain for mobile or edge devices, ensuring lightweight processing is a fundamental sustainability constraint.

1.3 Professional and Ethical Issues

Due to the collection and processing of extremely sensitive biometric data, the BioAuth system's development entails substantial professional and ethical obligations. The ACM Code of Ethics and Professional Conduct and the Software Engineering Code of Ethics and Professional Practice (IEEE Computer Society) serve as tight guidelines for the design and execution of this project.

The safe and open management of facial and physiological data is a fundamental ethical requirement that is consistent with the ACM Code's privacy principle. Continuous facial surveillance is necessary to extract rPPG signals, which has inherent privacy problems. In order to solve this, the system enforces ephemeral data processing; biometric inputs are specifically forbidden from being permanently saved or sent without informed user agreement, and they are examined in volatile memory for liveness detection.

Additionally, according to the IEEE Code of Ethics, maintaining justice and refraining from prejudice are fundamental professional obligations. Because BioAuth depends on optical signal extraction (e.g., CHROM and POS algorithms), differences in user skin tone, facial structure, and ambient illumination might have a disproportionate impact on performance. In order to reduce algorithmic bias and avoid systemic disadvantages across various demographic groups, it is our ethical duty as engineers to assess the system using representative and diverse datasets.

Honest reporting of the system's capabilities and limits is another need of professional integrity. A false sense of security in crucial deployment environments (like financial technology) could result from overstating the liveness detection's resilience against sophisticated deepfakes.

Lastly, all vulnerability testing and spoofing simulations (such replay or injection attacks) must be strictly limited to isolated, controlled environments utilizing approved datasets in order to comply with professional conduct. It is illegal and unethical to use experimental attack vectors against live, real-world platforms.

2. Requirements

The BioAuth system's functional and non-functional components are outlined in the following specifications. These specifications guarantee the system's dependability, security, and efficiency in real-world remote authentication scenarios.

2.1 Functional Requirements

To determine whether the identity being given is that of a living person, the BioAuth system must perform biometric authentication by combining physiological and visual analysis.

- **Face Detection and Tracking:**

  Using camera input, the system will identify and continually monitor a human face in real time. Reliable signal extraction is supported by accurate tracking, which guarantees consistency across frames.
- **Facial Landmark Extraction:**

  In order to precisely choose regions of interest (ROI), the algorithm will recognize important face landmarks. In order to isolate facial regions appropriate for physiological signal processing, these markers are crucial.
- **Physiological Signal Extraction (rPPG):**

  By spatially averaging RGB values from the isolated ROIs (such as the forehead and cheeks) and using specific algorithms like CHROM or POS, the system will extract raw rPPG signals. In order to separate the human heart rate frequency band (e.g., 0.7 Hz to 4.0 Hz), the resulting data must be processed through bandpass filters.
- **Liveness Detection:**

  By computing the Signal-to-Noise Ratio (SNR) of the extracted pulse wave in the

frequency domain, the system will assess physiological liveness. A liveness failure will be triggered by signals that lack a clear, physiologically convincing frequency peak (such as those that indicate a paper mask or computer screen).

- **Face Similarity Verification:**

  In order to ensure consistency, the system will use deep learning architectures (such as ArcFace or DeepFace) to extract facial embeddings and calculate cosine similarity or distance metrics against the reference identity.

- **Motion and Quality Validation:**

  To ascertain whether the recorded data is appropriate for dependable processing, the system will examine motion, lighting, and overall input quality. It is important to recognize and deal with low-quality inputs.

- **Multi-Factor Decision Mechanism:**

  To make a final authentication judgment, the system will incorporate findings from motion validation, physiological signal analysis, and face likeness.

- **Authentication Output:**

  Authentication attempts will be categorized by the system as Accept, Reject, or Retry, with Retry being utilized when input quality is inadequate for a trustworthy determination.

- **Logging and Evaluation:**

  To facilitate assessment and system enhancement, the system must log authentication results and pertinent performance metrics.

2.2 Non-Functional Requirements

2.2.1 User Interface Requirements

The BioAuth system must provide a clear and intuitive interface to ensure smooth interaction during authentication.

- **Real-Time Camera Feedback:**

  The system shall display live camera input, allowing users to adjust positioning and environmental conditions.

- **Clear User Guidance:**

  The system shall provide real-time, context-aware feedback (e.g., 'Low Illumination

Detected' or 'Excessive Head Movement') to ensure the captured data meets the minimum threshold for reliable rPPG processing.

- **Status Indication:**
  The system shall clearly indicate system states such as processing, success, failure, or retry.
- **Result Explanation:**
  The system should provide brief explanations for unsuccessful authentication attempts (e.g., low lighting, unstable signal).
- **Non-Intrusive Interaction:**
  The system shall operate without requiring complex or unnatural user actions.

2.2.2 Performance Requirements

The system must operate efficiently to provide a responsive and reliable authentication experience.

- **Real-Time Processing:**
  The system shall process the live camera feed at a minimum of 30 Frames Per Second (FPS) to ensure sufficient temporal data points are collected for accurate rPPG signal extraction and frequency analysis.
- **Low Latency:**
  The system shall render the final authentication decision (fusion of face similarity and liveness score) within 3 to 5 seconds of continuous, stable face capture, ensuring a frictionless user experience without compromising security.
- **Efficient Resource Usage:**
  The system shall operate effectively on standard devices without excessive computational overhead.
- **Stability:**
  The system shall maintain consistent operation without crashes or interruptions.

2.2.3 Security and Privacy Requirements

The BioAuth system must ensure secure handling of biometric data and prevent misuse.

- **Secure Data Processing:**
  The system shall process biometric data securely, preventing unauthorized access.
- **No Unnecessary Data Storage:**
  The system shall avoid storing sensitive biometric data unless explicitly required.
- **User Consent:**
  The system shall require explicit user consent before processing biometric data.
- **Spoofing Resistance:**
  By evaluating physiological liveness via rPPG, the system shall detect and resist presentation attacks (e.g., print, replay) and synthetic deepfake inputs.
- **Data Protection Compliance:**
  The system should follow general data protection principles, including transparency and responsible data usage.

2.2.4 System and Deployment Requirements

The system must be deployable in realistic environments using commonly available technologies.

- **Cross-Platform Compatibility:**
  The system should support both web and mobile platforms.
- **Standard Hardware Usage:**
  The system shall operate using standard cameras without requiring specialized equipment.
- **Scalability:**
  The system should allow future extensions and improvements.
- **Ease of Deployment:**
  The system should be deployable with minimal setup and configuration effort.

3. References

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